09 · Does the "€100 -> Gold tier" perk retain people? — RDD (CausalPy)

July 12, 2026

> **Runs in the `legacy` environment (pymc<6): `make env-legacy`, kernel `cmp-legacy`.**

The business decision. Customers who cross **€100** annual spend get **Gold** status and its perks. Gold members retain better — but of course they do, they're big spenders. Does the *perk itself* boost retention, or are we just seeing that big spenders would retain anyway? And where should we set the threshold?

0.0.1 The idea: a sharp rule creates a natural experiment

Regression discontinuity (RDD) exploits the fact that Gold status is assigned by a **sharp rule** on a **running variable** (annual spend): below €100 you're out, at €100-or-above you're in. Now compare a customer at €99.50 with one at €100.50. On *retention potential* they are essentially identical — nobody is meaningfully more loyal for having spent one euro more. The *only* thing that differs between them is that one got the perk and one didn't. So **any jump in retention exactly at €100** must be caused by the perk, not by the underlying "big spenders retain more" trend (which is smooth through the cutoff). RDD reads off the size of that jump.

0.0.2 What it can and can't tell you

- It estimates a **LATE at the cutoff** — the perk's effect *for customers near €100* (here *local* means local-to-the-cutoff: an ATE exactly at €100, **not** the complier-LATE of a fuzzy design). It says nothing about your €500 whales; don't extrapolate the jump to them.
- It's only valid if customers can't **manipulate** their spend to just clear €100 (that would make the people just above and just below systematically different). We test this with a **simplified McCrary-style** density check (a symmetric-window binomial test for a suspicious pile-up of customers just above the cutoff; the full McCrary 2008 test fits a local-linear density with a Wald SE).
- A headline RD number is only as good as its **robustness checks**: does the jump survive different **bandwidths** (how wide a window around €100 we use), **placebo cutoffs** (fake thresholds should show no jump), and **polynomial order** (linear vs curved fits)? We run all of them.

> **On real data.** RDD fits anywhere a **threshold rule** assigns a treatment: loyalty tiers, credit-score cutoffs for an offer, "spend €X for free shipping," scholarship/eligibility scores. You need the running variable, the cutoff, and the outcome — all of which live in your CRM.

It follows the repo's fixed **7-step contract**: question -> simulate -> identify -> estimate -> validate -> decide in € -> caveats.

0.1 2 · Simulate a ground truth

Retention rises smoothly with annual spend (big spenders retain more anyway) — **plus** a true **+14pp jump** exactly at €100 from the Gold perk. The smooth part is the confounder; the jump is the causal effect. Spend is concentrated around the cutoff so both sides are well-populated.

The data-generating model — exactly what `dgp.rdd_perk` implements (defaults & seed in `src/cmp/dgp.py`). $n = 2000$ customers, cutoff $c = 100$:

$$S \sim \mathcal{N}(100,\, 40^2) \text{ (clipped below at 1)}, \qquad D = \mathbf{1}[S \geq 100],$$
$$p(S) = \operatorname{clip}\Big(\underbrace{0.45 + 0.0015\,(S-100)}_{\text{smooth baseline}} + \underbrace{0.14\,D}_{\text{the jump}},\ 0.02,\ 0.98\Big), \qquad Y = \text{retained} \sim \text{Bernoulli}(p(S)).$$

The smooth term $0.0015\,(S-100)$ is "big spenders retain more anyway" — the trend that fools the naive Gold-vs-non-Gold comparison but passes *continuously* through the cutoff (it is deliberately **linear**, so a local-linear fit is unbiased and the robustness section can vary bandwidth and polynomial order against a clean benchmark). The $0.14\,D$ term is the perk's true +14pp causal jump. Centering spend on the cutoff ($S \sim \mathcal{N}(100, 40^2)$) keeps both sides of the threshold densely populated.

```
TRUE perk effect = +14% retention at €100  ·  50% are Gold

        spend  treated  retention
0   84.325206        0        1.0
1  101.996816        1        0.0
2   93.392393        0        1.0
3   93.281224        0        0.0
4  102.504635        1        0.0
```

0.1.1 2b · First, watch the naive comparison fail

The obvious analysis — the one every loyalty dashboard already shows — is *average retention of Gold members vs everyone else*. Before fixing it, let's compute it and see exactly **how** wrong it is. Because we wrote the DGP, we can decompose the naive gap in closed form. Write $\Delta = S - c$ (spend centered on the cutoff) and $D = \mathbf{1}[\Delta \geq 0]$; then

$$\underbrace{\mathbb{E}[Y \mid D{=}1] - \mathbb{E}[Y \mid D{=}0]}_{\text{naive Gold-vs-rest gap}} = \underbrace{0.14}_{\text{true jump}} + \underbrace{0.0015\,(\mathbb{E}[\Delta \mid \Delta \geq 0] - \mathbb{E}[\Delta \mid \Delta < 0])}_{\text{selection: Gold members are big spenders}}.$$

With $\Delta \sim \mathcal{N}(0, 40^2)$, each one-sided mean is $40\sqrt{2/\pi} \approx 31.9$ in absolute value, so the selection term is $0.0015 \times 2 \times 31.9 \approx +9.6$pp (spend and probability clipping make this slightly approximate). Roughly **+10pp of pure selection** stacked on top of the +14pp causal jump — the naive gap credits the perk with the whole "big spenders retain anyway" trend. That +10pp is exactly what RDD will subtract off by comparing customers only *at the margin*, where the selection term vanishes.

```
Naive Gold-vs-rest gap: +23.9%   =   true causal jump +14%   +   selection bias
+9.9%
(closed-form prediction of the selection term: +9.6%; the small remainder is
clipping + sampling noise)
The naive dashboard number nearly doubles the perk's true effect - pure 'big
spenders retain anyway' selection.
```

0.2 3 · Identify — a jump at the cutoff is causal (and its validity conditions)

Potential outcomes. Let $Y_i(1)$ and $Y_i(0)$ be customer i's retention *with* and *without* the perk, S_i the running variable (annual spend), and $D_i = \mathbf{1}[S_i \geq c]$ the sharp assignment rule at the cutoff c (here €100). The estimand RD targets is the perk's effect **for customers at the margin**:

$$\tau_{RD} \;=\; \mathbb{E}[Y(1) - Y(0) \,|\, S = c].$$

Assumption (continuity). The conditional means $x \mapsto \mathbb{E}[Y(d) \mid S = x]$ are *continuous at c* for $d \in \{0,1\}$. In words: **nothing else that drives retention jumps at €100** — loyalty, tenure, customer quality all pass smoothly through the threshold; the perk is the only thing that switches. Then the observed jump identifies τ_{RD} in two steps:

$$\begin{aligned}
&\lim_{x\downarrow c} \mathbb{E}[Y \mid S{=}x] - \lim_{x\uparrow c} \mathbb{E}[Y \mid S{=}x] \\
&\quad= \lim_{x\downarrow c} \mathbb{E}[Y(1) \mid S{=}x] - \lim_{x\uparrow c} \mathbb{E}[Y(0) \mid S{=}x] \\
&\quad= \mathbb{E}[Y(1) \mid S{=}c] - \mathbb{E}[Y(0) \mid S{=}c] \;=\; \tau_{RD}.
\end{aligned}$$

First equality: just above the cutoff every customer is treated, so the observed Y *is* $Y(1)$; just below, it *is* $Y(0)$. Second equality: continuity lets each one-sided limit converge to its value *at* c. So the observable jump in average retention at €100 **equals** the causal effect at €100 — no unconfoundedness, no propensity model; the sharp rule plus continuity does all the work. The price is *locality*: this is a **LATE at the cutoff** — an ATE exactly at €100, silent about your €500 whales.

Manipulation is a continuity violation — and it leaves fingerprints. If customers (or account managers) sort across the threshold — nudging a €97 basket to €101 — the *kind* of customer just above €100 differs from the kind just below, so $\mathbb{E}[Y(d) \mid S = x]$ becomes *discontinuous* at c and the derivation collapses. Continuity itself is untestable (it's about potential outcomes), but sorting leaves observable fingerprints, which gives us two **testable implications**: the *density* of spend should be smooth at €100 (the McCrary check, 3b), and *pre-determined covariates* should be smooth at €100 (the balance check, 3c). They can't prove continuity; they can catch it failing.

Sharp vs fuzzy — where this design sits. Here the rule is *sharp*: $P(D{=}1 \mid S{=}x)$ jumps from 0 to 1 at €100. When the rule is leaky (grandfathered members, manual overrides), crossing the threshold only *raises the probability* of Gold, and the estimand becomes the **fuzzy-RD** ratio

$$\tau_{\text{fuzzy}} \;=\; \frac{\lim_{x\downarrow c} \mathbb{E}[Y \mid S{=}x] - \lim_{x\uparrow c} \mathbb{E}[Y \mid S{=}x]}{\lim_{x\downarrow c} \mathbb{E}[D \mid S{=}x] - \lim_{x\uparrow c} \mathbb{E}[D \mid S{=}x]}$$

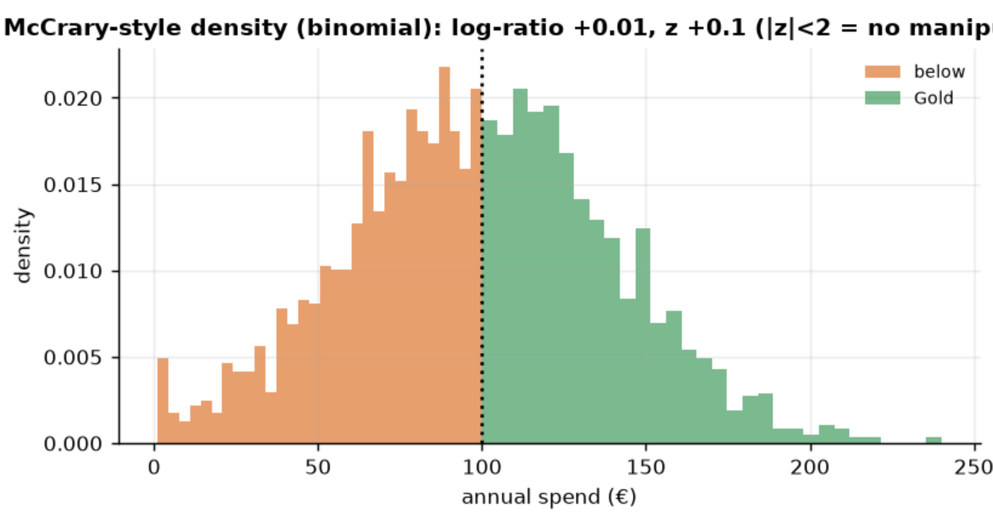
— the outcome jump rescaled by the treatment-probability jump. That ratio is exactly a **Wald/IV estimator** with "crossed the threshold" as the instrument (notebook 11), and it identifies a *complier*-LATE at the cutoff: the effect for customers whose Gold status actually flips when they cross €100. In our sharp design the denominator is 1 and τ_{fuzzy} collapses to τ_{RD} — which is the precise sense in which "local" here means local-to-the-cutoff, *not* complier-local. Two things must hold for the sharp analysis, both checked below:

- **No manipulation** of the running variable at the cutoff — the McCrary-style density check (3b) plus covariate continuity (3c).
- **The functional form isn't driving the jump** — bandwidth, placebo cutoffs, and polynomial order (Step 5).

0.3 3b · Validity check 1 — McCrary-style density check (manipulation)

If customers game their spend to just clear €100 (or the business nudges them), the *density* of spend jumps at the cutoff and RD breaks. We compare the density just below vs just above with a symmetric-window binomial test. (This is a *simplified* version; the full McCrary 2008 test fits a local-linear density on each side with a Wald SE.) This is the first of the two testable implications of continuity derived in Step 3; the second — pre-determined covariates must not jump — follows in 3c.

```
density just below 23.7 vs above 24.0, log-ratio +0.01, binomial z +0.15 - no
significant sorting across the cutoff (|z| < 2).
```



(The density is smooth through €100 — no suspicious pile-up of customers *just* above the cutoff — so we have no evidence of manipulation, and the RDD comparison is fair.)

0.3.1 3c · Validity check 2 — pre-determined covariates don't jump at the cutoff

Continuity has a **second** testable implication. Anything determined *before* the perk decision — how long the customer has been with us, how many orders they placed last year — *cannot* be caused by it. So if we run the very same RD machinery with a **pre-determined covariate as**

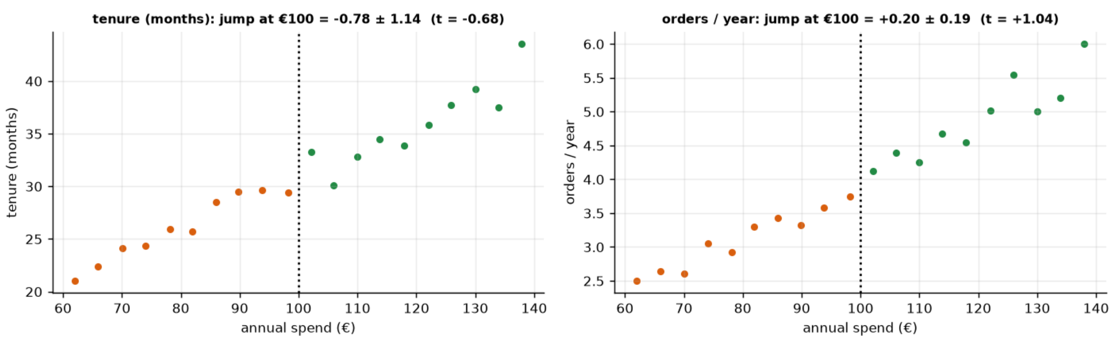
the "outcome", the estimated jump at €100 must be ~ 0. A real jump in, say, tenure would mean the *people* just above and just below the threshold are different people — continuity violated — exactly like an unbalanced covariate in a botched RCT, localized to the cutoff. Together with McCrary this is the standard falsification *pair* of applied RD (the Lee–Lemieux checklist).

Our simulated CRM has no covariates, so we create two that mirror the real-world pattern: strongly **correlated with spend** (so they'd fool a naive comparison) but **smooth through €100 by construction** (nothing in how they're generated knows about the cutoff — the null is true, and the test should say so). On real CRM data you run this on *every* pre-treatment column you have and report the whole table, like an RCT balance table.

To run the check — and later the whole Step-5 robustness battery — we need a fast, reusable **local-linear RD fit**: plain OLS with a standard error on the jump, so every check is judged by a t-statistic rather than an eyeball.

```
pre-determined covariate tenure_m : jump at €100 = -0.78  (t = -0.68)


pre-determined covariate n_orders : jump at €100 = +0.20  (t = +1.04)
Both |t| < 2 - covariates pass smoothly through the cutoff; the people either
side are comparable.
```



How to read it. Both covariates climb steeply with spend — Gold members *are* longer-tenured, heavier orderers, which is precisely why the naive comparison in 2b lied — but each trend passes **smoothly** through €100: the local-linear jump is statistically zero for both (t-stats printed above). Combined with the clean McCrary density, this is the RD analogue of an RCT balance check: near the threshold, *who you are* doesn't change — only Gold status does. Had tenure jumped at €100, no amount of modelling would rescue the design; the fix is to find out *why* (account managers gaming the tier? a billing process that rounds spend up?) — not a fancier regression.

0.4 4 · Estimate — Bayesian regression discontinuity

Now the estimate itself. We fit a **local linear** regression on each side of the cutoff — a straight line for the below-€100 customers and another for the Gold customers — and the **height of the step where they meet at €100** is the causal effect of the perk. We restrict to a **bandwidth** of +/-€40 around the cutoff: the customers near the threshold are the comparable ones (a €98 and a €102 customer are alike; a €500 whale is not). Wider bandwidths borrow non-comparable customers; narrower ones throw away data — Step 5 checks the answer is stable to that choice. In production you wouldn't hand-pick the window: use a plug-in optimal bandwidth (Imbens–Kalyanaraman 2012, or the Calonico–Cattaneo–Titiunik procedure behind `rdrobust` in R/Python), and treat the Step-5 sweep as the check that the choice doesn't matter.

The fitted model, in symbols (within the bandwidth $|S_i - 100| \le 40$, with $D_i = \mathbf{1}[S_i \ge 100]$):

$$\text{retention}_i \sim \mathcal{N}(\mu_i,\ \sigma^2), \qquad \mu_i = \beta_0 + \beta_1 S_i + \tau D_i + \beta_2\, S_i D_i,$$

$$\beta_0,\ \beta_1,\ \tau,\ \beta_2\ \sim\ \mathcal{N}(0,\, 50^2), \qquad \sigma\ \sim\ \text{HalfNormal}(1)$$

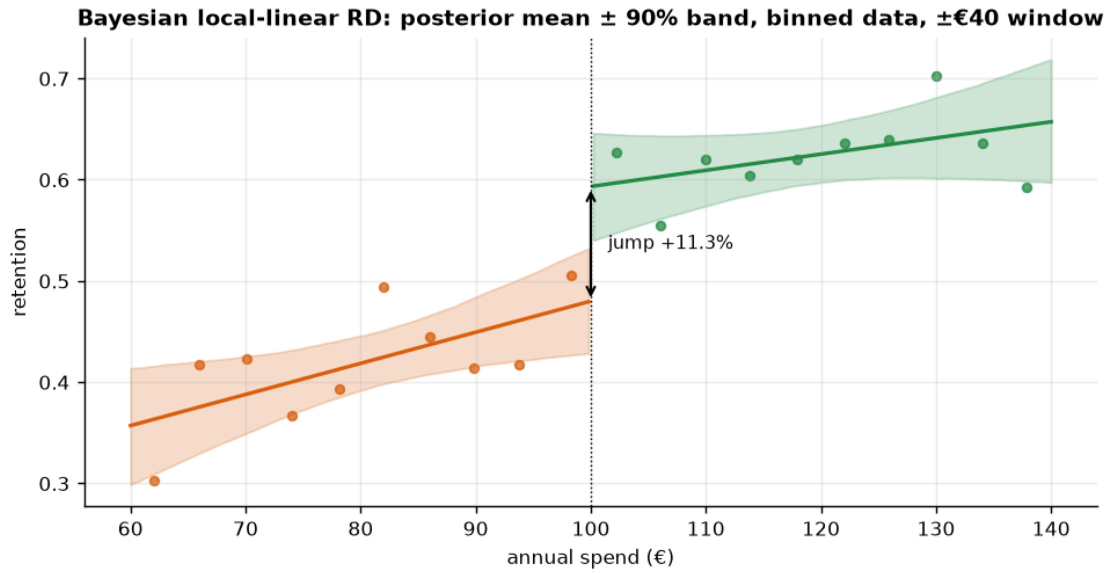
— one straight line per side (intercepts β_0 vs $\beta_0 + \tau$, slopes β_1 vs $\beta_1 + \beta_2$; the formula in the code, `retention ~ 1 + spend + treated + spend:treated`, is this equation). Because spend is *uncentered*, the causal jump is **not** the raw τ coefficient: it is the gap between the two fitted lines evaluated *at the cutoff* — CausalPy's `discontinuity_at_threshold`, the difference of predictions just above vs just below €100.

The priors are CausalPy's defaults — and here that's fine. The two prior lines above are exactly what CausalPy's `LinearRegression` samples (transcribed from its source: `beta ~ Normal(0, 50)` on every design-matrix coefficient, `y_hat_sigma ~ HalfNormal(1)` on the noise — the `NUTS: [beta, y_hat_sigma]` line in the sampler log — visible when you run the notebook — is those two blocks). On this problem they are effectively flat: retention lives in $[0, 1]$, so every plausible coefficient is orders of magnitude smaller than the prior sd of 50, and a Bernoulli outcome forces $\sigma \le 0.5$, comfortably inside HalfNormal(1). The posterior is therefore data-dominated — appropriate for a local fit whose one job is to measure a jump, and it's why the Bayesian estimate and the OLS approximations of Step 5 should (and do) agree. But note that "the defaults are harmless" is a statement about **units**, not a law: score retention in percentage points (0–100) or spend in cents and you'd want to rescale or rethink the priors.

Owning the linear-probability choice. Retention is Bernoulli, so the Normal likelihood is misspecified twice: the true noise variance is $p(1-p)$ (heteroskedastic — it moves with p), and a straight line isn't confined to $[0, 1]$. Within a +/-€40 window, though, p moves only a few points, so $p(1-p)$ is nearly constant and the fitted lines stay far from the boundaries — the misspecification is second-order, and in exchange τ reads *directly* in probability points, the currency of Step 6's economics. The honest check is to refit with a logit link and confirm the implied jump at the cutoff barely moves — we do exactly that right after the headline fit.

```
RD jump +11.3% (true +14%)  ·  90% credible interval [+3.8%, +18.7%]
RDD convergence: max r-hat 1.000 - min ESS 1378 - divergences 0
```

```
The height of the step where the two fitted lines meet at €100 IS the estimate:
+11.3% (90% CrI [+3.8%, +18.7%]).
Contrast: the naive Step-2b gap (+23.9%) compared the average heights of the two
clouds instead.
```



This is the picture the whole method lives in. Each dot is the average retention of a €4-wide spend bin inside the +/-€40 window; the two lines are the posterior mean of the local-linear fit on each side, with 90% bands. **The height of the step where the two lines meet at €100 *is* the estimate**, and the bands' separation at the cutoff is its uncertainty. Contrast this with the naive comparison of 2b: that compared the *average height* of the orange cloud against the green cloud, crediting the perk with the entire upward trend; RD reads only the vertical break at the threshold, so the trend cancels. Also read the bands honestly: they are visibly wide relative to the step — that width, not the point estimate, is what will drive the business call in Step 6.

And the promised **LPM sanity check**: refit the same window with a logit link (a likelihood that *is* Bernoulli) and compare the implied jump in retention probability at €100.

```
Jump at €100 - linear-probability model: +11.3% (Bayesian posterior mean)  ·
logit link: +11.2% - the link choice is immaterial inside a narrow window, as
argued in Step 4.
```

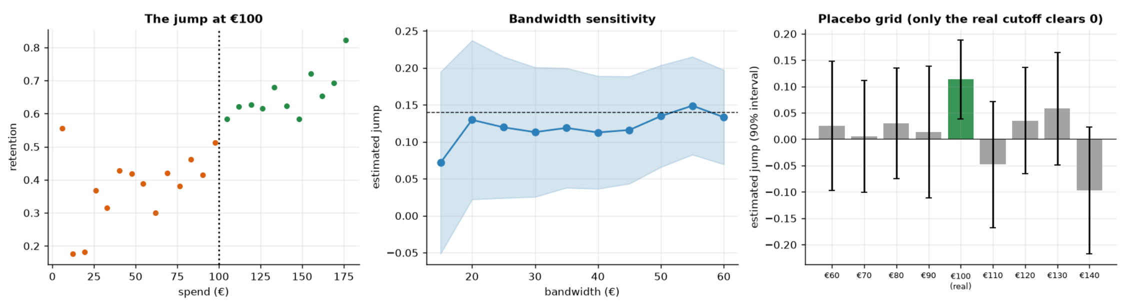
0.5 5 · Validate — see the jump, plus bandwidth / placebo / polynomial robustness

A credible RD number is one that **survives** the researcher-degrees-of-freedom checks:

- **Bandwidth curve** — the estimate should be stable across a sensible range of bandwidths.
- **Placebo cutoffs** — run RD at a *grid* of fake thresholds (€60–€90 and €110–€140) where there is no perk; every estimated jump should be ~ 0. A real jump at a fake cutoff would mean the method finds discontinuities in noise. Construction detail: below-€100 placebos use only non-Gold customers and above-€100 placebos only Gold customers, so no placebo window straddles the real jump — the standard one-sided placebo construction.
- **Polynomial order** — linear vs quadratic local fit shouldn't move the answer much. We deliberately stop at quadratic: Gelman & Imbens (2019) showed *high-order global* polynomials manufacture spurious jumps (noisy, boundary-dominated weights), which is exactly why

modern RD practice is local-linear within a bandwidth rather than a global polynomial fit.

```
Bandwidth sweep: +7.2% … +14.9% across bw 15-60 (mean +12.0%) - drifts mildly
but stays clearly positive.
Polynomial: linear +11.3%, quadratic +10.9%.
Placebo grid (jump, t): €60 +2.5% (t=+0.34), €70 +0.6% (t=+0.09), €80 +3.0%
(t=+0.47), €90 +1.4% (t=+0.18), €110 -4.8% (t=-0.66), €120 +3.5% (t=+0.58), €130
+5.8% (t=+0.90), €140 -9.7% (t=-1.33)
Largest placebo |t| = 1.33 (at €140) - all 8 placebos statistically
indistinguishable from zero; the real €100 jump's 90% CrI [+3.8%, +18.7%]
excludes zero.
```



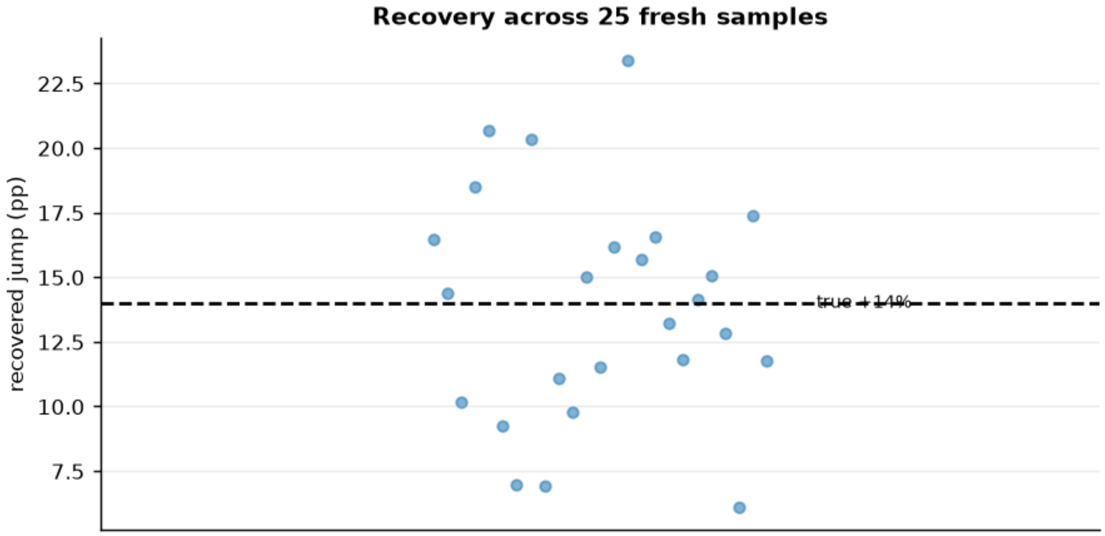
How to read the robustness battery. *Left* — bin retention by spend and you can *see* the step at €100 sitting on top of the smooth "big spenders retain more" trend; that step is the perk effect. *Middle* — the estimate **drifts mildly** with the bandwidth (narrow windows are noisier, wider ones pull in curvature) but stays clearly positive and brackets the truth across the whole range, so we're not cherry-picking a window. *Right* — the decisive check, now a **grid of eight fake cutoffs with 90% intervals**: re-run the same local-linear RD at €60–€90 (non-Gold customers only) and €110–€140 (Gold only), where no perk switches. The point estimates wobble around zero — one or two bars always *look* sizeable (with eight draws from noise, they should!), but read each bar's *interval*, not its height: every placebo interval comfortably spans zero and every $|t|$ is under 2, with the largest flagged in the printout so you can check it rather than trust us. Only the **real** €100 cutoff has an interval that **excludes** zero. That is the honest test — a *family* of t-stats, not an eyeball. Together with the McCrary density, the covariate-continuity check, and the linear-vs-quadratic agreement, it's the full case that the jump is a real causal effect, not an artefact of how we drew the lines.

(Implementation note: for speed, the battery's refits are local-linear ***OLS approximations*** *of the same specification, judged on frequentist t-stats; the headline €100-cutoff estimate itself stays fully Bayesian.)*

0.5.1 5b · Recovery across many seeds — unbiased *and* calibrated?

A single sample can land a little off; the real test is whether the estimator is **centred on the truth over repeated samples** and whether its 90% interval **covers** the truth at the stated rate. We refit on many fresh samples (a small fast fit each) and check both.

```
RD jump across 25 seeds: mean +13.8% (true +14%)  bias -0.2%  sd 4.3%  ·  90% CI
covers truth in 23/25 seeds - near-nominal calibration.
```

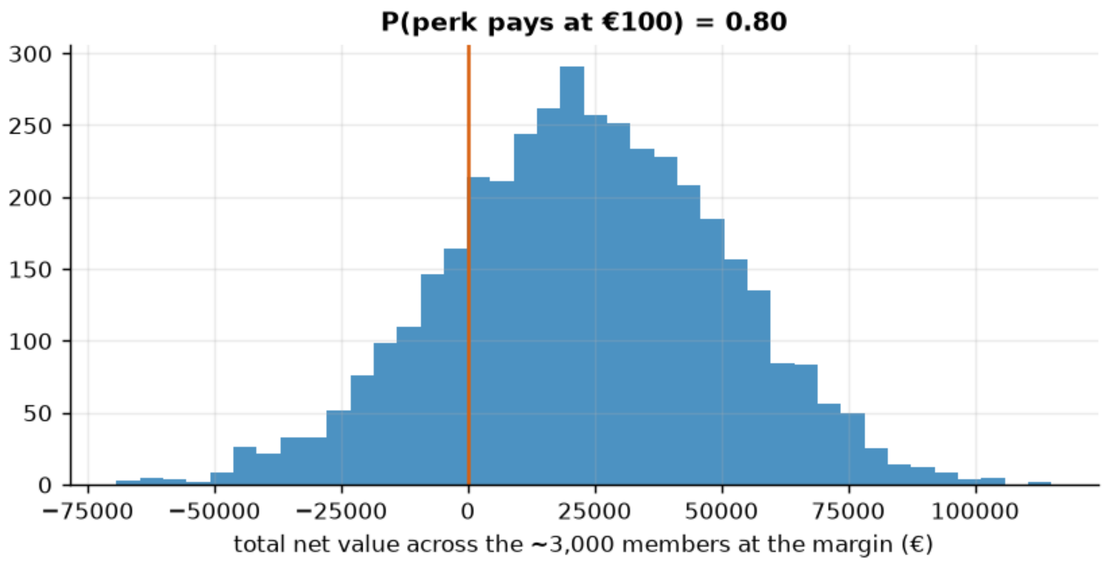


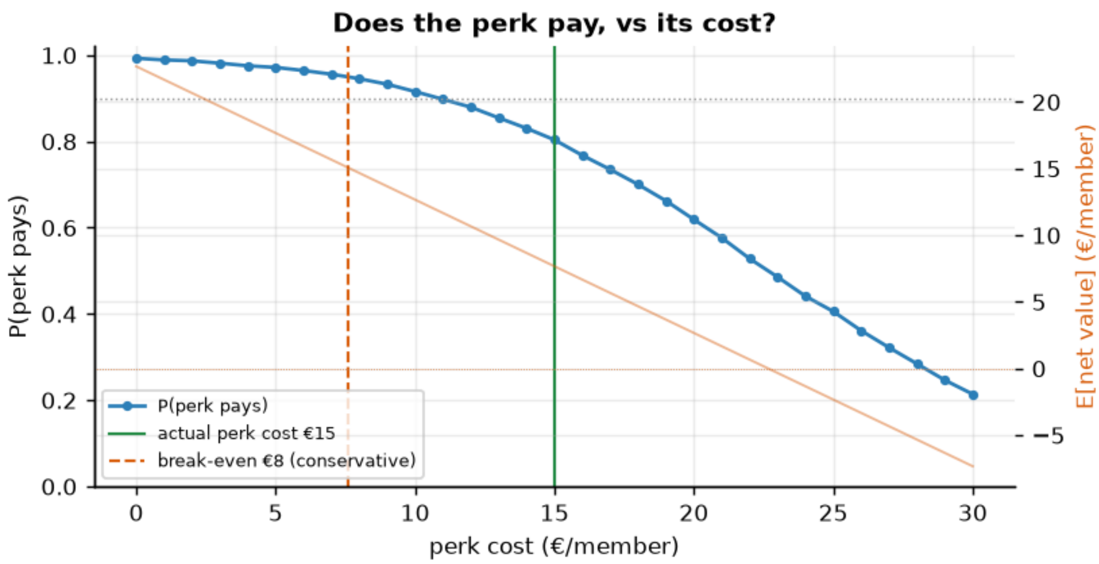
0.6 6 · Decide, in euros — does the perk pay, and where to set the cutoff?

Value the retention lift: extra retained customers x annual value, minus the perk cost, **at the current €100 margin** (that's what RD identifies). We also sweep the perk cost to find the break-even.

> **What RD can and can't answer.** RD identifies *only* the local effect at the €100 margin — it tells us whether the perk pays **there**, but it **cannot** tell us where to set the cutoff (moving the threshold extrapolates beyond the RD's local validity; that question needs an experiment or multi-cutoff data).
>
> **Two stated inputs.** `N_AT_MARGIN ~ 3,000` is an illustrative count of members inside the +/-€40 window around €100 per cycle — swap in your CRM's number; the **0.9** go/no-go bar is the cookbook's convention for "sure enough to act without a further test".

```
Net €7.7/member  ·  total €22,960 [90% €-22,140, €67,444]
P(perk pays) 0.80 -> reconsider  ·  break-even perk cost ~ €8/member
(conservative).
Perk-cost sweep €0-30: P(perk pays) falls 0.99 -> 0.21; at the actual €15 cost
P(pays) = 0.80, E[net] = €8/member (break-even €8).
```





How to read the decision histogram. Mass to the right of the orange zero line = P(perk pays); the wide spread — *not* the positive mean — is what makes this a *reconsider / TEST* call.

And the sweep turns it into a negotiation. At the actual €15 perk cost, P(pays) comes out around 0.8 (printed above) — real but below the 0.9 bar — while the *conservative* break-even (the cost the 5th-percentile effect still covers) sits in the €6–8/member range (exact value printed above). So the decision isn't "keep or kill": it's **either bring the perk's cost toward the break-even** (a cheaper perk with the same salience keeps the retention jump and clears the bar by construction) **or buy more certainty before renewing** — a longer measurement horizon or a wider eligibility window around the cutoff, both of which tighten the jump posterior. What the sweep must *not* be used for is moving the €100 threshold itself: RD's evidence is local to the customers near €100 (the blockquote above), and pricing a new cutoff from this posterior would be extrapolation dressed as analysis.

How much certainty would more data actually buy? The next cell quantifies it — the last input the memo needs.

```
Members in the +/-€40 window this cycle: 1,706  ·  P(perk pays €15) now: 0.80
One more cycle of data (~ 2x the window sample): CrI ~ 29% tighter -> P(pays) ~
0.88 if the mean holds.
Clearing the 0.9 bar at the current cost needs posterior sd €9.1 -> €6.0/member,
i.e. ~ 2.3x the window sample - which is why renegotiating the perk's cost is
the stronger lever.
```

0.6.1 The one-paragraph decision

> **To the CMO.** The Gold perk causes a real retention lift for customers at the €100 margin — a jump of roughly ten percentage points (exact posterior printed in Step 4), an estimate that survives a density check, covariate continuity, eight placebo cutoffs, bandwidth and polynomial sweeps, and a multi-seed calibration study. At the current €15/member cost it pays *in expectation* (order €20k per cycle across the ~3,000 marginal members), but the probability it pays is only ~ 0.8 — below our 0.9 action bar. So: **don't kill it, and don't blindly renew it.** Either (a) renegotiate the perk's cost down toward the conservative break-even printed in Step 6 — the cost even the 5th-percentile effect covers, which clears the bar by construction — or (b) buy certainty: the value-of-information readout above shows one more cycle of margin data tightens the interval ~30% and lifts P(pays) toward the bar *if the point estimate holds*, while clearing 0.9 outright at the current cost would take several times the data. Cost renegotiation is the stronger lever; more data is the fallback.
>
> **And the threshold question the title poses?** RD alone **cannot** answer "where should the threshold be": it measures the effect at €100 and nothing in the data distinguishes a constant jump from one that decays away from the cutoff — formally, $\tau(s)$ is identified only at $s = c$ (our DGP happens to make it constant; a real perk needn't be). The design answer: **randomize the cutoff** — a tier experiment assigning €90 / €100 / €110 thresholds across regions or cohorts traces $\tau(s)$ at three points and prices the threshold directly; failing that, a past threshold change gives a second cutoff for a multi-cutoff RD. Moving the threshold on the strength of one local estimate would be extrapolation dressed as analysis.

0.7 7 · Caveats

- **RD is *local*.** The effect is identified only *at the cutoff* — the perk's effect on customers near €100, not on your €500 whales. Don't extrapolate the jump to everyone.
- **Manipulation breaks it.** We checked the McCrary density and covariate continuity; watch for heaping at round numbers. If the observations *nearest* the threshold are suspect (heaping, gamed baskets), the standard remedy is a **donut RD** — drop observations within +/-eps of the cutoff and refit (CausalPy exposes this as `donut_hole`).
- **Bandwidth is bias–variance.** The estimate drifts mildly with the window (narrow = noisier, wide = more curvature) but stays clearly positive here; an estimate that *swings*

across zero with bandwidth would be a red flag.

- **Fuzzy RD if the rule isn't sharp.** If crossing €100 only *raises the probability* of Gold (grandfathering, overrides), use the fuzzy-RD ratio (jump in Y ÷ jump in P(treated)) — this ratio is exactly the Wald/IV estimator of notebook 11, with the threshold acting as the instrument (estimand formalized in Step 3).